THE BASE CHANGE OF AN INTEGRAL RING MAP ALONG A FIBRE PRODUCT IS INTEGRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The Situation of a fibre product of rings, with the identification of the two kernels of the structure maps. situation at source lines 521-542 and the discussion of the same section. Supplies the rings and the coincidence of the two kernels used in the adjustment.
- Integrality of a ring map, defined by the existence of a monic polynomial relation with coefficients in the source for every element of the target. used at proof lines 613-616. Provides the polynomial satisfied by the chosen element.

Source reference. Original lemma 0E1S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17582–17582. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10885–10885. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In Situation 08KH if $B \to A$ is integral, then $B' \to A'$ is integral.*

Proof. Let $a' \in A'$ with image $a \in A$. Let $x^d + b_1 x^{d-1} + \ldots + b_d$ be a monical polynomial with coefficients in B satisfied by a. Choose $b'_i \in B'$ mapping to $b_i \in B$ (possible). Then $(a')^d + b'_1(a')^{d-1} + \ldots + b'_d$ is in the kernel of $A' \to A$. Since

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$\mathrm{Ker}(B' \to B) = \mathrm{Ker}(A' \to A)$ we can modify our choice of b'_d to get $(a')^d + b'_1(a')^{d-1} + \ldots + b'_d = 0$ as desired. $\square$